The halogens and noble gases

THESE ARE GROUPS 17 AND 18 OF THE PERIODIC TABLE.

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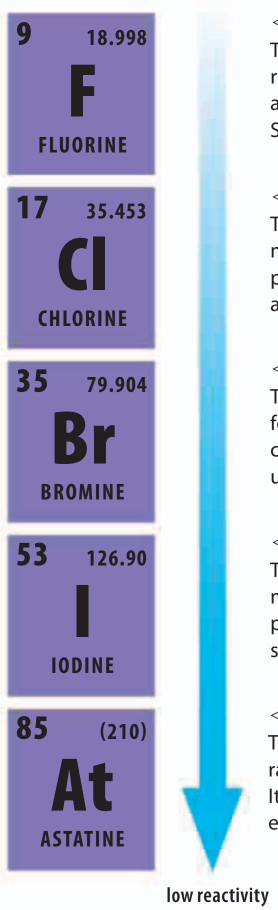
While the left side of the periodic table is dominated by metals, the right side—made up of Groups 17 and 18—are all nonmetals. The chemical characteristics of these two groups could not be more different.

The halogen group

There are five naturally occurring halogens. The reactivity decreases down the group as the atoms grow larger. The outer shell of a smaller atom is closer to the nucleus, so the electrons in it are held more strongly than in larger ones—and this includes the electron added during reactions to make an ion.



◁ **Fluorine**
This pale yellow gas is the most reactive nonmetal element of all, and so forms compounds easily. Sodium fluoride is found in toothpaste.

◁ **Chlorine**
This is a green gas that is used in many disinfectants and cleaning products, such as bleach. Chlorides are added to many swimming pools.

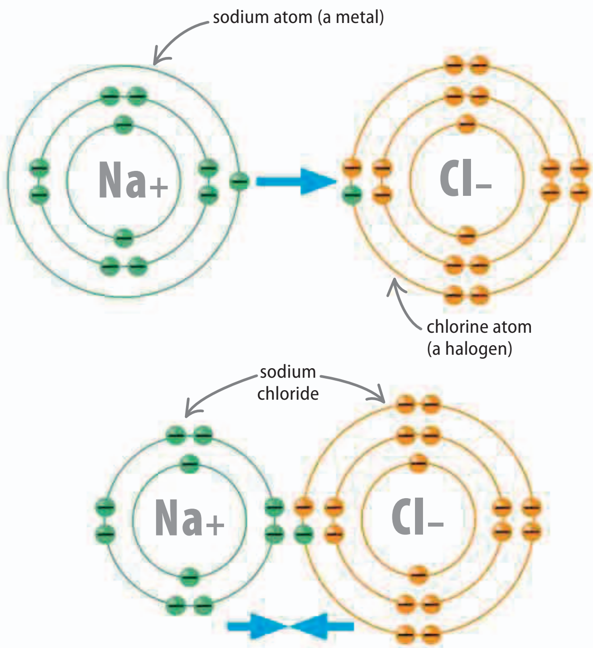
◁ **Bromine**
This is the only nonmetal element found in liquid form in standard conditions. Its compounds are used in fireproofing.

◁ **Iodine**
This purple-gray solid does not melt into a liquid at atmospheric pressure; it changes from a solid state straight into a purple gas.

◁ **Astatine**
The heaviest halogen is highly radioactive and is very rare. Its atoms break up into other elements quickly.

The salt formers

The members of Group 17 are also known as the halogens, meaning "salt formers." The atoms of halogens have outer shells with seven electrons—out of a maximum of eight. As a result, all the halogens are very electronegative, meaning that they form negatively charged ions easily by attracting an electron each to fill their outer shells. They do this by reacting with metals (which form positively charged ions) to form stable ionic compounds. These substances are called salts.



△ **Common salt**
It is perhaps no surprise that the most common salt is called common salt or table salt. This is a compound formed when the halogen chlorine (Cl) reacts with the metal sodium (Na), producing sodium chloride (NaCl).